A Technology for Assisting Literacy Development in Adults with Dyslexia and Illiterate Second Language Learners



Matteo Cristani, Serena Dal Maso, Sabrina Piccinin, Claudio Tomazzoli, Marco Vedovato, and Maria Vender

Abstract Developing good literacy skills is a domain in which two distinct populations of adults display marked difficulties: individuals suffering from developmental dyslexia and second language learners. These weaknesses in literacy can have a negative impact on their quality of life and in particular on their employability and professional carrier. These subjects share, however, advanced *digital skills*, in particular those related to the usage of mobile devices. Leveraging on this skills we developed a platform that allows several different subjects (teachers, project managers, learners) in interacting with each other for assisting the development of the above mentioned literacy proficiency.

Keywords Dyslexia · Language learning · Second language

1 Introduction

The project aims at carrying out accurate investigations and training for strengthening the profile and employment prospects of two populations on the Italian territory for which access to reading and writing is problematic: adults suffering from developmental dyslexia (dyslexia henceforth) and individuals who, due to their migration

M. Cristani (✉) · C. Tomazzoli · M. Vedovato
Dipartimento di Informatica, Università degli Studi di Verona, Verona, Italy
e-mail: matteo.cristani@univr.it

M. Vedovato
e-mail: marco.vedovato@studenti.univr.it

S. D. Maso · S. Piccinin · M. Vender
Dipartimento di Culture e Civiltà, Università degli Studi di Verona, Verona, Italy
e-mail: serena.dalmaso@univr.it

S. Piccinin
e-mail: sabrina.piccinin@univr.it

M. Vender
e-mail: maria.vender@univr.it

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2021

V. L. Uskov et al. (eds.), *Smart Education and e-Learning 2021*, Smart Innovation, Systems and Technologies 240, https://doi.org/10.1007/978-981-16-2834-4_40

background, have learned Italian as a second language, possibly with a starting condition of first language illiteracy (either full, namely with no reading or writing skills, or partial as in those subjects able to read and write in their L1 language, but unable to do so in L2). Although there are many investigations that have already been carried out in the past on this theme, these needs impact in a novel era, where the widespread diffusion of mobile technologies has brought potential of leverage in the above mentioned populations. Within the project has therefore taken space the idea that it would be useful to support in this way learning and also assistance to learning, control of progress and analyses of the outcomes as driven by the project's concepts. The platform that has been designed, and prototyped in Flask, is illustrated here. The research team, formed by linguists and computer scientists, has devised a model based on the up-to-date methods for the mentioned cases and implemented them in a system able to support various steps of the learning process. The model of continuous progress control is validated in the current literature on acquisition of reading skills in the mentioned populations, and it is devised to support the activities in potential further other groups with similar literacy deficits, but coming from further specifically limited contexts. The solution that we propose does not exist as a means for supporting learning activities in adults with dyslexia or second language illiterate adults. There exist a few solutions for children with dyslexia, and some of these have been adapted to work for adults, whilst for illiterate second language learners this is the first one to be deployed. Moreover, the technology is developed as a prototype exactly as a base for users' feedback by the adults, that are indeed in the experimental perimeter. The technology is developed with a line towards the expression of the ability of users to anticipate the learning processes determined by elements of a specific exercise. In particular, during a training, a teacher or a language specialist could identify limits to the developed exercise and evolve it to satisfy the requests of specific types of users (in particular this holds for second language learners with specific geographic provenance).

The rest of the paper is organised as follows: Sect. 2 discusses the theoretical framework introduced in the project. We then provide a general description of the application in Sect. 3 and specify the functional architecture of the prototype in Sects. 3.1 and 3.2. Section 4 introduces some related work and Sect. 5 takes some conclusions and sketches further work.

2 Theoretical Framework

Adopting a narrow interpretation, literacy refers to the ability to read and write through an automatized process, usually acquired with formal education and schooling. This automatism requires the development and the systematic recourse, during reading, to a lexical route (i.e., global visual representation of the lexical item), in addition to the phonological route (i.e., grapheme-phoneme decoding). The second skill is generally used for unknown or unfamiliar words by skilled readers [1]. This decoding ability is often referred to as "primary literacy", opposed to "functional

literacy", which, in a wider sense, has to do with the ability to deal with texts, understanding their meanings, communicative goals and interactions with the contexts. Therefore, successfully reading a text involves the acquisition of a range of complex language structures and the ability to process those structures in real time. Such competences include the awareness of linguistic sounds (phonology) and their relation to graphemes and spelling patterns (orthography), word meaning (semantics), morpho-syntactic structures and textual organization. An automatized processing of all these competences is the necessary condition to develop reading fluency and text comprehension. During the last decades, a large scientific literature has been developed on what should be meant by literacy and how to deal with the problem of assessing literacy skills in education and other social contexts, including assistive learning. In most recent research, a growing emphasis has been put on the application of literacy skills in everyday life. Based on the above mentioned viewpoint the research community has developed a concept of measure on the literacy skill and on their measure by means of continuous scales rather than in terms of dichotomies (literates vs. illiterates). Although dealing with texts in many daily tasks and work routines is extremely common, it remains an obstacle for specific sections of population, both in terms of understanding and production of written texts. This is particularly the case for two specific populations, which will be targeted in our project: (a) Native speakers of Italian with dyslexia and (b) non-native speakers of Italian (i.e., migrants, refugees, asylum seekers) with absent or low literacy skills in their second language. The main characteristics of these two populations, both suffering from reading and spelling impairments which are however related to different conditions, will be now briefly discussed. Dyslexia is a lifelong condition characterized by reading and spelling deficits [2], together with impairments in the phonological [3, 4], morphological [5, 6] and grammatical competence [7, 8]. These difficulties have been found to persist also in adulthood [9], with negative consequences on the individual's academic and professional careers [10]. Despite representing a rather different type of population, non-literate and low-literate learners of L2 Italian experience similar difficulties in the area of basic literacy skills. Such learners face the double challenge of having to learn another language, while simultaneously trying to grasp the alphabetic principle underlying its writing system. The difficulties encountered by these individuals can locate at rather different levels, because of the highly heterogenous nature of their profile. These may span from learners still struggling with the understanding of the basic functions of print to those who, being literate in a writing system other than alphabetic, exhibit an inadequate degree of phonemic awareness [11], a skill which is instrumental for cracking the alphabetic code [12, 13]. As social and work-related demands increase continuously, reading and writing skills which would have been considered adequate in the past are often no longer sufficient to gain and maintain employment. Moreover, the current labor market is characterized by both an increase of automatization in the workplace and a higher demand for literacy skills, even for the most manual jobs. Indeed, only few jobs can be performed successfully if employees have problems with reading and writing. The general acknowledgment that a low literacy degree is likely to have negative effects at the level of employability has given rise to a number of programs supporting adult education and literacy in developing

countries (EFA Programs, 'Education For All', supported by UNESCO, is one of these). However, Europe is also confronted with a growing challenge for a wider and more equal access to literacy skills and should aim at a more inclusive continuing educational system. So far, the landmarks provided by the Council of Europe and by LESLLA (Literacy Education Second Language Learning for Adults) for both educational policy and research have rarely been integrated into actual intervention guidelines for employability improvement. As a consequence, governmental institutions are not fully resourced to face these new challenges and lack specific tools aiming to support the development of the more solid literacy abilities that are likely to enhance employability. Within the project, some of the experimental activities have been conceived in a way that is strongly supported, in the literacy proficiency, by the usage of the envisioned technology. In particular one important activity shall be the literacy training, which could represent an important opportunity for both individuals with dyslexia and non-native speakers with low literacy skills. Each experimental group is assigned to a specific intervention program, which is developed based on the profiles and on the identified specific learning needs. Literacy skills will be trained by means of specific tools aimed at guiding, supporting and automatizing a fast and accurate recognition and consolidation of syllables, morphemes and single words, and partly therefore by the usage of the platform documented in this work. Both the sub-lexical and the lexical routes for reading [1] are trained, in order to enhance automaticity and reduce reading efforts, while in parallel also reinforcing writing skills. Morphological awareness will also be specifically trained, as both populations have shown sensitivity to the structure of morphologically complex words during visual word recognition tasks [14, 15] and it has been proved to be particularly effective for the rehabilitation of reading and writing disorders [16–18]. Analogously, in the project there will be a specific pursue of literacy training for non-native speakers of Italian.

3 The Application

In the platform we designed and prototyped the core concept is the *project*. A project is a complex activity that involves a project manager, who launched the project itself, and two further groups of users: the *operators* and the *learners*. The users are all managed by the *application managers*. When an instance of the platform is deployed onto a server, an application manager is the one who installed it. She has no specific power different from other application managers who are the further users she can register. When someone involved in a specific learning project asks for deploying it, the application manager registers the new user as a project manager, and therefore gives her the privileges needed to create new projects. The application managers have also the privilege of making a project manager an application manager too. When a project is created with its own description, then the project manager can provide the content of the project itself by acting on the registration of operators, who are in charge of managing the learning activities. In particular, the operator registers

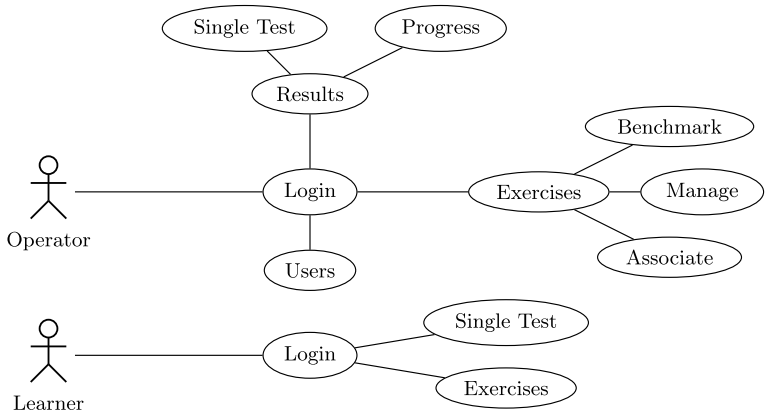


Fig. 1 Use case operator/learner

new *exercises*, that in turn shall be mounted into specific sequences and groups, that constitute the model of deployment for the application, that we describe in details in Sect. 3.1. The exercise constitute the basic brick of the notion of project and the model of learning underlying the application concept. Every exercise is formed by a *task* that is assigned to a learner, a set of *objects* to be displayed on the tool interface, that can be either *texts*, *videos*, *images* or *sounds*. On each of the above objects the operator can add *tags*, that are associated to answers, and that are displayed in form of clickable segments of the rendering. Executing an exercise shall be either one *single click* or a *sequence of clicks*, or a *selection of elements by clicks*, or an input on a further text *input box*. The operator mounts an exercise and the learner executes the exercise by reading the task, and then executing the operations required. To each exercise are associated more than one admissible answers. The operator associates an *evaluation method* for the exercise (either *autonomously* or *twin*, that means that the value of the exercise is assigned by the operator when the learner executes the exercise). To each answer of one exercise, the operator associates a value that is assigned automatically, and can be corrected by the operator when the exercise is in twin modality only. Every execution of an exercise is logged in terms of duration. Every learner registration is under the supervision of a single operator. Once an operator has added a specific learner to a project, the learner is involved in the project itself. On creation, the operator assigns to the learner a *level* of knowledge of the host language (in the specific case, Italian). A learner can perform exercises of a level that is lower than or equal to hers.

3.1 Application Functions

In this section we address the matter of practical functions of the target application, leaving out the managing tasks for the sake of space. We provide a descriptive UML Use Case in Fig. 1 where the use case Manage stands for the basic CRUD (Create, Read, Update, Delete) persistence operations (Fig. 2).



Fig. 2 Two screenshots of the application

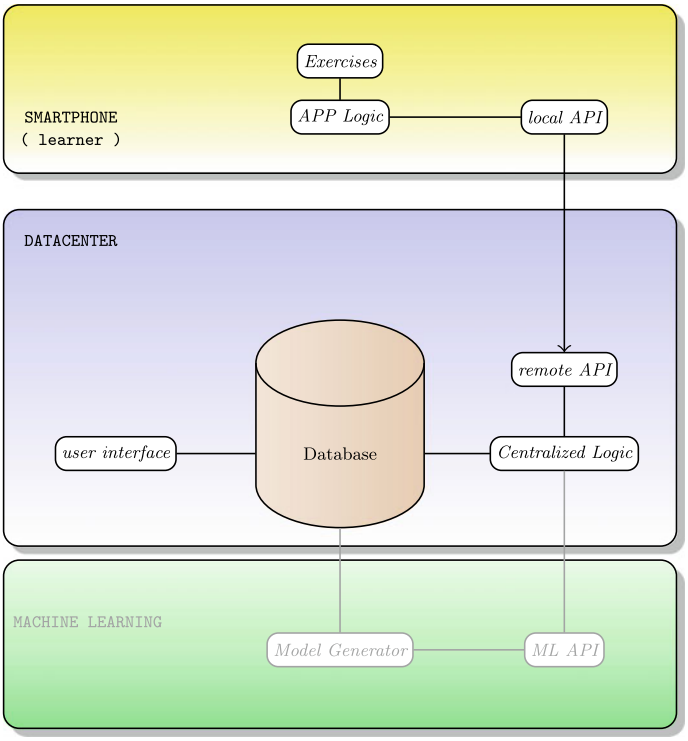


Fig. 3 Logical model of the application

3.2 Application Architecture

The application is a quite common standard *app* whose logical model is described in Fig. 3. Three main layers are described along with their modules: the main interface for *learners* will be responsive. The managing functions are organized in modules which are stored on the cloud. On the server, data are collected and on a background process are passed to a machine learning algorithm used to identify the patterns in common, and use these patterns for the evaluation of progresses.

4 Related Work

Although experimental studies reporting results of reading intervention in adults with dyslexia are still rather sparse, there is evidence indicating that their difficulties in decoding, fluency and comprehension can be successfully compensated for if they are provided with a specific training. The results of a recent systematic review [19] seem to indicate that phonological and morphological interventions, which are typically

beneficial for enhancing reading accuracy, could be integrated with a computerized intervention aiming at improving fluency more specifically. The reading acceleration training, in particular, has provided promising results indicating that imposing time constraints on reading by means of computerized programs can be very effective for automatizing and speeding up the process of decoding, with positive gains extending to reading comprehension as well. Similarly, scientifically validated evidence on effective literacy training for non-literate and low-literate L2 learners is rather scant, as highlighted by a recent review on the topic [20], with most classroom-based studies only providing an impressionistic account of strategies which may be useful with such learners [21]. For languages with a fairly transparent orthography, such as Italian, phonics-based methods, focusing on recognizing and automatizing grapheme-phoneme associations while gradually moving to syllables and words, are generally thought to be among the effective strategies to teach reading (especially if such methods are contextualized, as suggested in [22]). However, a major problem with these methods is represented by the huge amount of instruction hours learners need in order to crack the alphabetic code. Teacher-fronted courses cannot typically allocate many hours to such activities, since this would subtract valuable time to the development of their L2 oral skills, a core need for learners trying to resettle in a new country, and indeed another factor which can boost literacy skills [23]. Even with small groups of students, the time the teacher can dedicate to each learner is inevitably very limited. Moreover, phonics-based activities often turn out to be highly demotivating for adult learners, who already exhibit unusually high dropout rates [23]. Providing students with a tool which promotes autonomous learning would therefore allow teachers to concentrate on interaction rather than focus on time-consuming phonics-based activities, which might be instead more successfully supported by technology, as showed by recent research on technology-assisted learning for LESLLA learners [24]. Furthermore, technology-assisted learning, if properly designed, can guarantee constant and automatic corrective feedback, a practice which has proved to prevent learners from automatizing their errors [25]. Given that such feedback is provided privately, anxiety that learners typically suffer from when corrected in front of their classmates can also be kept at bay, thus creating optimal learning conditions. The application model is inspired by studies by some of the authors on the topic of message deployment in social networks [26, 27], on the topic of multimodal structure of web applications [28, 29] and on the structure of negotiation systems [30, 31]. Technology support to people with dyslexia is a long-term effort since the half of the years 2000 [32]. In 2010, a systematic review of the literature has shown the success (few) and the failures (many) the investigations have reached then [33]. The years between 2010 and 2016 as the most fruitful ones in the study of technological solutions for adults with dyslexia and the learning of language. In particular, some important milestones have been obtained with ‘Dyslexia baca’ [34], an application that has been quite successful and employed in a number of randomised studies thenceforth, and EasyLexia [35], also quite a valid solution essentially obtained in the same period of time. Since then, many studies have been proceeding further. Recently, a new review on the difficulties of these application has been proposed [36] that is concerned with the ability of the adults with dyslexia to work on the web

and mobile devices. It should be also noticed that young children and adults are out of the regular spectrum of technology implementations and only a few methods, such as those cited above have been based upon solutions with this type of feature. Even more limited is the set of studies on illiterate adults' language learning technologies. The very first attempt that is documented in literature back to 2010 [37] and is part of a long-lasting project on the topic. Further on, a more modern approach to the problem is discussed in [38]. Some thought on the principles of design that have to be used in this context are discussed in [39] that have been a reference for the present work.

5 Conclusions and Further Work

In this paper we have discussed the development of an application employed to devise and employ exercises to reinforce the proficiency in language reading and writing for adults with dyslexia and illiterate second language learners. The application is developed as a prototype within a wider intervention for language skill enhancement for employability improvement. The application has shown to be functional for the mentioned purpose from an analytical viewpoint and the prototype is validated in a group of experts, formed by twenty teachers with some training in adults with dyslexia. These have provided a direct feedback on functionalities that reads on a score 8.2 on a scale of 10. One of the major strengths of this research lies in the synergy established between information technology and psycholinguistics and in the integration of different domains of expertise for the development of an original intervention program aimed at enhancing reading skills of people with fragile literacies that can also improve their employability and, more in general, their quality of life. Further steps of this investigation are two: (a) we shall employ and therefore test the application in the real world situations provided as interventions in the project, (b) we shall improve the application in order to achieve higher level quality for further implementations, including the extension to meta-linguistic feedbacks.

Acknowledgements Authors gratefully thank Fondazione Cariverona, for support to this research under "Bando Ricerca Scientifica di Eccellenza 2018" All authors contribute equally to this work.

References

1. Coltheart, M., Rastle, K., Perry, C., Langdon R., Ziegler, J.: Drc: a dual route cascaded model of visual word recognition and reading aloud. Psych. Rev. (2001)
2. Vellutino Frank, R., Fletcher, J.M., Snowling, M.J., Scanlon, D.M.: Specific reading disability (dyslexia): what have we learned in the past four decades? J. Child Psychol. Psych. Allied Discip. **45**(1), 2–40 (2004)
3. Ramus, F., Pidgeon, E., Frith, U.: The relationship between motor control and phonology in dyslexic children. J. Child Psychol. Psych. Allied Discip. **44**(5), 712–722 (2003)

4. Vender, M., Delfitto, D., Melloni, C.: How do bilingual dyslexic and typically developing children perform in nonword repetition? Evidence from a study on Italian l2 children. Bilingualism **23**(4), 884–896 (2020)
5. Joanisse, M.F., Manis, F.R., Keating, P., Seidenberg, M.S.: Language deficits in dyslexic children: Speech perception, phonology, and morphology. J. Experim. Child Psychol. **77**(1), 30–60 (2000)
6. Vender, M., Mantione, F., Savazzi, S., Delfitto, D., Melloni, C.: Inflectional morphology and dyslexia: Italian children's performance in a nonword pluralization task. Ann. Dyslexia **67**(3), 401–426 (2017)
7. Vender, M., Hu, S., Mantione, F., Delfitto, D., Melloni, C.: The production of clitic pronouns: a study on bilingual and monolingual dyslexic children. Front. Psychol. 9 (2018)
8. Wiseheart, R., Altmann, L.J.P., Park, H., Lombardino, L.J.: Sentence comprehension in young adults with developmental dyslexia. Ann. Dyslexia **59**(2), 151–167 (2009)
9. Bruck, M.: Persistence of dyslexics' phonological awareness deficits. Development. Psychol. **28**(5), 874–886 (1992)
10. Gerber, P.J.: The impact of learning disabilities on adulthood: a review of the evidenced-based literature for research and practice in adult education. J. Learn. Disabil. **45**(1), 31–46 (2012)
11. Young-Scholten, M., Strom, N.: First-time l2 readers: is there a critical period? In: Kurvers, J., van de Craats, I., Young-Scholten, M., (eds.) Low Educated Adult Second Language and Literacy Acquisition: Proceedings of the Inaugural Conference, Utrecht, pp. 45–68 (2006)
12. de Gelder, B., Vroomen, J., Bertelson, P.: The effects of alphabetic-reading competence on language representation in bilingual Chinese subjects. Psychol. Res. **55**(4), 315–321 (1993)
13. Read, C., Yun-Fei, Z., Hong-Yin, N., Bao-Qing, D.: The ability to manipulate speech sounds depends on knowing alphabetic writing. Cognition **24**(1–2), 31–44 (1986)
14. Burani, C., Marcolini, S., Traficante, D., Zoccolotti, P.: Reading derived words by Italian children with and without dyslexia: the effect of root length. Front. Psychol. **9** (2018)
15. Piccinin, S., Dal Maso, S., Giraudo, H.: Bound stem processing in l1 and l2 Italian. Lingue e Linguaggio **17**(2), 289–305 (2018)
16. McArthur, G., Eve, P.M., Jones, K., Banales, E., Kohnen, S., Anandakumar, T., Larsen, L., Marinus, E., Wang, H.C., Castles, A.: Phonics training for english-speaking poor readers. Cochrane Database System. Rev. **12**(CD009115) (2012)
17. Goodwin, A.P., Ahn, S.: A meta-analysis of morphological interventions: effects on literacy achievement of children with literacy difficulties. Ann. Dyslexia **60**(2), 183–208 (2010)
18. Bowers, P.N., Kirby, J.N., Deacon, S.H.: The effects of morphological instruction on literacy skills: a systematic review of the literature. Rev. Educ. Res. **80**(2), 144–179 (2010)
19. Maria, V., Melloni, C., Delfitto, D.: The effectiveness of reading intervention in adults with dyslexia: a systematic review (under review)
20. Piccinin, S., Dal Maso, S.: Promoting literacy in adult second language learners: a systematic review of effective practices (under review)
21. Bigelow, M., Schwarz, R.L.: Adult English Language Learners with Limited Literacy. National Institute for Literacy, Washington, DC (2010)
22. Vinogradow, P.: Balancing top and bottom: learner generated texts for teaching phonics. In: Wall, T., Leong, M. (eds.), Low-Educated Second Language and Literacy Acquisition: Proceedings of the 5th Symposium, pp. 3–14. Bow Valley College, Calgary (2010)
23. Condelli, L., Wrigley, H.S., Yoon, K.S.: The what works study: instruction, literacy and language learning for adult ESL literacy students. In: Reder, S., Bynner, J. (eds.) Tracking adult literacy and numeracy: longitudinal studies of adult education, pp. 132–159. Routledge, New York (2009)
24. Eva, M.: From computer-assisted to technology-enhanced learning. lessons learnt and fast forward toward (digital) literacy of LESLLA learners. In: D'Agostino, M., Mocciaro, E. (eds.) Languages and Literacy in New Migration. Research, Practice, and Policy. Literacy Education and Second Language Learning for Adults (LESLLA): Proceedings of the 14th Annual Meeting of LESLLA, Palermo, 4–6 Oct 2018. UniPa Press, Palermo (2020)

25. Lyster, Roy, Saito, Kazuya, Sato, Masatoshi: Oral corrective feedback in second language classrooms. Lang. Teach. **46**(1), 1–40 (2013)
26. Cristani, M., Fogoroasi, D., Tomazzoli, C.: Measuring Homophily, vol. 1748 (2016)
27. Cristani, M., Tomazzoli, C., Olivieri, F.: Semantic social network analysis foresees message flows. **1**, 296–303 (2016)
28. Cristani, M., Tomazzoli, C.: A multimodal approach to exploit similarity in documents. **8481**, 490–499 (2014)
29. Cristani, M., Tomazzoli, C.: A multimodal approach to relevance and pertinence of documents. LNCS **9799**, 157–168 (2016)
30. Burato, E., Cristani, M.: Learning as meaning negotiation: a model based on english auction. LNCS **5559**, 60–69 (2009)
31. Burato, E., Cristani, M.: The process of reaching agreement in meaning negotiation. LNCS **7270**, 1–42 (2012)
32. Woodfine, B.P., Nunes, M.B., Wright, D.J.: Text-based synchronous e-learning and dyslexia: not necessarily the perfect match!. Comput. Educ. **50**(3), 703–717 (2008)
33. McCarthy, J.E., Swierenga, S.J.: What we know about dyslexia and web accessibility: a research review. Universal Access Inform. Soc. **9**(2), 147–152 (2010)
34. Daud, S.M., Abas, H.: 'dyslexia baca' Mobile App—The Learning Ecosystem for Dyslexic Children, pp. 412–416 (2013)
35. Skiada, R., Soroniati, E., Gardeli, A., Zissis, D.: Easylexia: A Mobile Application for Children with Learning Difficulties, vol. 27, pp. 218–228 (2014)
36. Kadam, P., Thaker, M., Vyas, G., Vegesna, A.: A learning ecosystem for dyslexic. In: EAI/Springer Innovations in Communication and Computing, pp. 29–39 (2021)
37. Lumsden, J., Leung, R., D'Amours, D., McDonald, D.: Alex: a mobile adult literacy experiential learning application. Int. J. Mob. Learn. Organ. **4**(2), 172–191 (2010)
38. Tirmizi, S.A.U., Iftikhar, Y., Ali, S., Ehsan, A., Ehsan, A., Shahid, S.: Ustaad: a mobile platform for teaching illiterates. LNCS **11747**, 788–796 (2019)
39. Mahmood, Z., Shahzadi, S.S., Tariq, S.: Content management and user interface for uneducated people. LNCS **8519**(PART 3), 432–441 (2014)